

Empowering Robot Designers: A Digital Tool for Early-Stage Social Robot Prototyping and Communication

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ABSTRACT

While social robots promise to benefit people with emotional and informational needs, persistent concerns exist around their ability to effectively manage expectations and meet user needs. In this study, a digital toolkit was developed as a possible solution to support the early-stage design of social robots and address the current absence of a structured, research-informed approach. The prototype toolkit provides an interactive and visual process to assist designers of different backgrounds and varying degrees of technical knowledge; and to guide the development team throughout iterative ideation, prototyping, and communication regarding new social robot designs. The toolkit employs established human-robot interaction principles and insights from existing research to incorporate a guiding framework in the design process. This paper presents key prototype features, an outline of the user flow, a preliminary expert evaluation of the tool's concept and usability, and future aims.

CCS CONCEPTS

• **Human-centered computing** → **User interface toolkits**; • **General and reference** → **Design**; • **Applied computing** → *Interactive learning environments*; *Computer-assisted instruction*.

KEYWORDS

Creativity Support, User Experience Design, HCI for Development, Prototyping

ACM Reference Format:

Matthew Milton, Shabnam Fakhrosseini, Chaiwoo Lee, and Joseph Coughlin. 2024. Empowering Robot Designers: A Digital Tool for Early-Stage Social Robot Prototyping and Communication. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3613905.3651048>

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3651048>

1 INTRODUCTION

Social robots have been developed for a range of applications, including entertainment, healthcare, and education, to meet complex social, emotional and physical needs of users. The social robot design space is complex and includes the functional task, social role, and physical embodiment of the robot [8]. While an array of social robot designs, functions, and features exists among research projects, relatively few robots are commercially available [20]. Additionally, their acceptance in settings such as homes, schools, and hospitals remain limited. Users adopt technologies for various reasons, including perceived ease of use and usefulness, as emphasized by Davis (1985) in his widely recognized technology acceptance model [7]. Previous research also suggests that a social robot's design, including its physical embodiment and features, impacts user trust and perception [8, 23, 26, 36].

The diversity of robots is partly mirrored in their designers. Social robot designers can have backgrounds in psychology, computer science, engineering, product design, industrial design, or related fields. Many operate within academic research labs or private companies with extensive robotics programs. This multidisciplinary group exhibits a range of technical knowledge. Accordingly, robot design teams have various challenges which can include cumbersome software design tools and difficulty communicating complex, domain-specific knowledge to others. Similarly, the technical nature of robot design may prove to be a barrier for incorporating end-users in the design process.

Is the development of social robots adequately considering end user needs? Are the values that end users associate with the use of social robots being evaluated? These questions, along with other user-centered concerns, should be considered at the early stages of product, engineering, and robot design processes [5, 15]. Participatory design methodology applied to Human-Robot Interaction (HRI) research attempts to support designers in understanding user perceptions and preferences [6, 18]. While methods exist for developing prototypes during the initial product design stages [10, 12], there is a lack of applied tools for generating ideas and designing social robots that align with users' needs and preferences. In this paper, we present a design tool that aims to assist designers with diverse levels of expertise to conceptualize and mock up early-stage social robot designs. Specifically, the tool focuses on conceptualizing low-fidelity designs of a robot's social capabilities and physical appearance rather than its algorithmic implementation or simulations.

2 RELATED WORK

2.1 Use of Social Robots and Related Design Challenges

Although definitions vary, social robots are often understood to be physically-embodied agents designed to interact with humans semi- or fully autonomously and in a social, natural, and interpersonal manner [2, 31]. To achieve human-like social skills, social robots are often equipped with an array of physical features along with cognitive and behavioral capabilities. Common design features include heads, faces, arms, wheels, LED screens, touch screens, lights, and hard external shells [20]. Social robots are often equipped with environmental sensors such as cameras, microphones, touch sensors, and lasers [20]. Beyond physical features, social robots are also able to recognize, interpret, and act on human social behavior. Common cognitive or behavioral capabilities include speech production, gesturing, facial expressions, head and eye movement, and autonomous navigation [20]. Social robots can come in different forms, including animal-like, humanoid, and mechanical appearances, among others. These characteristics allow social robots to serve in various roles and settings, such as education, entertainment, service, healthcare, research, and telepresence [20].

Despite widespread potential applications for social robots, their commercial availability is historically low [20]. In a review of 9920 articles, Mahdi et al. (2022) identified 344 robots for analysis across various dimensions, roughly 55% of which were research prototypes that never reached the market and only 28.6% (n=98) were commercially available [20].

The limited success of social robots may be attributed to a lack of a holistic design approach. Past studies suggested that thoughtful physical and behavioral design of social robots is crucial to successful adoption. Hameed et al. (2016) investigated factors that impact user adoption and acceptance of social robots, and indicated that a robot's function, appearance, and behavior must closely match [11]. For instance, a robot designed to resemble a human should behave like one as well. Similarly, Breazeal et al. (2016) suggested that social robots must be competent in verbal and non-verbal interactions, connect to users on a cognitive and emotional level, and behave in a manner that is consistent with user expectations and needs [2]. The physical embodiment of social robots is also known to mediate user interaction and trust [23, 26, 36]. For instance, a humanoid robot might elicit a sense of familiarity and trust, whereas a robot with an unconventional or unsettling appearance might generate skepticism or discomfort [23].

The design of social robots occasionally neglects a balanced examination of how users experience their physical, cognitive, and behavioral elements. For example, the Stretch teleoperation robot, which is designed to support user independence within the homes by helping with everyday domestic tasks [13], does not align with design guidelines stating that users of varying ages prefer personal robots with an unthreatening and calming presence, a friendly face, and a design that blends seamlessly into their homes [14]. Similarly, early versions of the functional capabilities of a social robot called CLARC were evaluated with older adults and clinicians, but an exploration of its aesthetics, embodiment, and shell design were delayed [32]. In the case of MyJay, a play robot which was initially prototyped at a high-fidelity without involving its primary end

users – caregivers and children, researchers discuss the intention to gather feedback from these groups as part of future work [21].

The earliest stages of product and engineering design are crucial. Nigel Cross (2000) writes in his book *Engineering Design Methods: Strategies for Product Design*, “The very first conceptualizations and representations of problem and solution are [...] critical to the kinds of searches and other procedures that will follow, and so to the final solution that will be designed” [5]. Some HRI research has identified human-centered design methodologies as particularly important to the early stages of robot design [6, 18, 34]. These methods involve introducing end-users to the robot prototypes early in development in order to gather their perceptions and needs [6]. Formative UX evaluations in HRI are especially useful in identifying user feedback on early robot designs. Linbolm et al. (2020) writes that feedback on “...conceptual ideas and low-fidelity prototypes from intended users provide necessary hints about the interaction quality, supporting selection of multiple design alternatives, and identifying UX problems and negative UX. It is easier and less expensive to change a robot design and interaction flow in the earlier phases of the development project than in the latter phases” [18]. Despite the recognized importance of early user involvement in HRI, there are few structured support tools specifically for social robot design teams to conceptualize robots and elicit end user needs and preferences.

2.2 Interactive Design Tools

Digital tools that support users in creating and visualizing complex designs exist in domains such as video games, consumer products, and interior design. For example, IKEA's Kreativ platform allows users to select a common type of room, search the IKEA product catalog, and place them in a 3D environment. Hovering over items with a mouse cursor reveals essential information, such as their price, availability, and customer reviews. IKEA Kreativ and other tools, such as those for product customization and avatar creation, provide a space for users of varying design skills and technical capabilities to explore creative design through a semi-structured workflow with intuitive controls, interactive information displays, and the ability to make and visualize changes.

Interactive early-stage design tools are less prevalent in the robot design space and typically pose barriers for new users, and designers and stakeholders without programming or human-robot interaction knowledge. For example, *Webots*, an open-source robot simulator, incorporates a 3D environment, a menu with a hierarchical representation of the simulated world, a source code editor, and a console for compilation and controller outputs in one interface [19] (see Appendix A.1). *CoppeliaSim*, also a multi-purpose simulation tool, has a complex user interface similar to *Webots* [4]. This complexity arises from a strong emphasis on technical simulation and detailed mechanical implementation, which requires complex programming and computing better suited for later stages of design. In addition, most existing platforms neglect cognitive and affective elements, and opportunities to explore zoomorphic or abstract design variations.

Therefore, there is a need for a more holistic and accessible approach to the design of social robots. This need arises from incorporating little-to-no user-informed data in the early stages of

social robot development on one hand, and the complexity of the process, which includes many factors within each aspect, on the other. Specifically, there is an opportunity to facilitate the collection of user feedback and to promote the integration of physical, social, and emotional design elements into the earliest stages of social robot development.

3 DEVELOPING A NEW SOCIAL ROBOT DESIGN TOOLKIT

3.1 Design Goals and Requirements

The high-level goals for our social robot design toolkit are as follows:

- (1) To provide social robot designers with a comprehensive, early-stage design tool that accounts for both physical and social-cognitive design considerations.
- (2) To create an accessible and low barrier-to-entry interface that enables interdisciplinary designers and their teams to contribute to the design discussion without a comprehensive or technical HRI background.
- (3) To provide a guided design process while leaving flexibility to explore new directions and applications.
- (4) To support the incorporation of end user needs in the robot design process.

The design toolkit is intended to support both experienced and inexperienced robot designers and foster rapid, creative work. We determined that design requirements for e-learning [1] and digital creativity platforms [29] were applicable and relevant goals. We have further divided relevant design requirements into high-level (or creativity support) and low-level (or interface design) requirements as presented in Appendix B.1.

3.2 Creating the User Flow

The user flow of the toolkit is divided into four major phases: Organize, Conceptualize, Visualize, and Share. This paper explores the Conceptualization and Visualization phases in detail.

The design flow developed by Deng et al. [8] was referenced as a foundation for the user flow of our social robot toolkit. Deng et al. (2019) suggest that social robot designers follow these steps: 1) Select the task the robot is intended to support, 2) Select the robot's social role, 3) Choose the robot embodiment and corresponding behaviors, and 4) Implement the robot. The authors also document and reveal patterns in the designs and embodiment of existing robots that could inform choices at each step in the process, allowing designers to reference previous robots that address similar tasks. With this work as a starting point, we introduce a sub-step for considering the cognitive capabilities of the robot (see Appendix A.2). In our toolkit, the first three steps in Deng's design flow plus our additional sub-step are called the Conceptualization Phase. Although the steps are initially presented in a linear fashion, the workflow can support a non-linear design approach through a navigation menu that enables designers to jump between steps and make adjustments in any order they choose.

In addition, our tool actualizes the implementation stage of the framework with a Visualization Phase. This phase is for rapidly

designing a low-fidelity 3D model of the robot following the Conceptualization Phase. The visualization can act as a useful frame of reference to be shared with other designers and could help imagine the robot in its intended context. Lastly, we included two design summaries (see Figure 1). One summary is generated at the end of the Conceptualization Phase and the other after the Visualization Phase. These summaries document the design choices a designer made throughout the process and are intended to facilitate collaboration, communication, and rapid feedback with other team members, including engineers, HRI experts, or end users.

3.3 Proposed Design Features and Options

The Conceptualization Phase boasts multiple screens with a recurring visual design including large cards that represent design options, and reveals detailed description and application examples upon hovering over them with a cursor. This design reduces the effort required to scan, navigate, and make decisions by minimizing the amount of information presented to the user at one time. The design also supports exploration and learning by encouraging designers to examine and read about the design choices available to them. Users are presented with one topic at a time (i.e., task selection, social role selection, etc.) yet their selections are always visible regardless of the page they are viewing. This feature aims to support designers to reflect on their design choices and allow them to quickly recognize their previous selections. Another feature is the suggested design choices which appear above the list of options on each page or as the starting position on a sliding scale. The suggestions are based on trends observed in existing social robot designs and more general HRI insights. Several comprehensive sources of social robot design data can serve as a reference for the presented design suggestions, including [8] and [9]. To provide relevant suggestions, the platform will assess the selections made by the designer throughout the Conceptualization Phase. The following describes detailed features of the proposed tool in the order which a user might interact with them.

3.3.1 Task Selection. Users are first asked to choose tasks that the robot would carry out. The tasks available to choose from are inspired by the Circumplex Model of Group Tasks [24] and some related to companionship (i.e., providing comfort, playing games). The Circumplex Model group tasks have been broken down into more specific tasks that are easier to interpret and apply. For example, for tasks that would be associated with 'planning' and 'performance/actions' we have included, 'provide light educational lessons' and 'help solve problems'. For demonstration, we have included the following additional tasks: 'clean room', 'provide comfort', 'converse', 'support decision-making', 'provide general information', 'provide reminders', and 'play games'. When users hover a cursor over an option, they can see the definition and example abilities the robot might need to accomplish the task (see figure 1).

3.3.2 Cognitive Capacities Selection. The tool will suggest the most relevant capacities to the selected tasks, which are informed by previous studies on user wants and perceptions of social robots [22, 27]. Alternatively, the designer can select capacities, then check the Task page for tasks requiring those capacities (see figure 1).

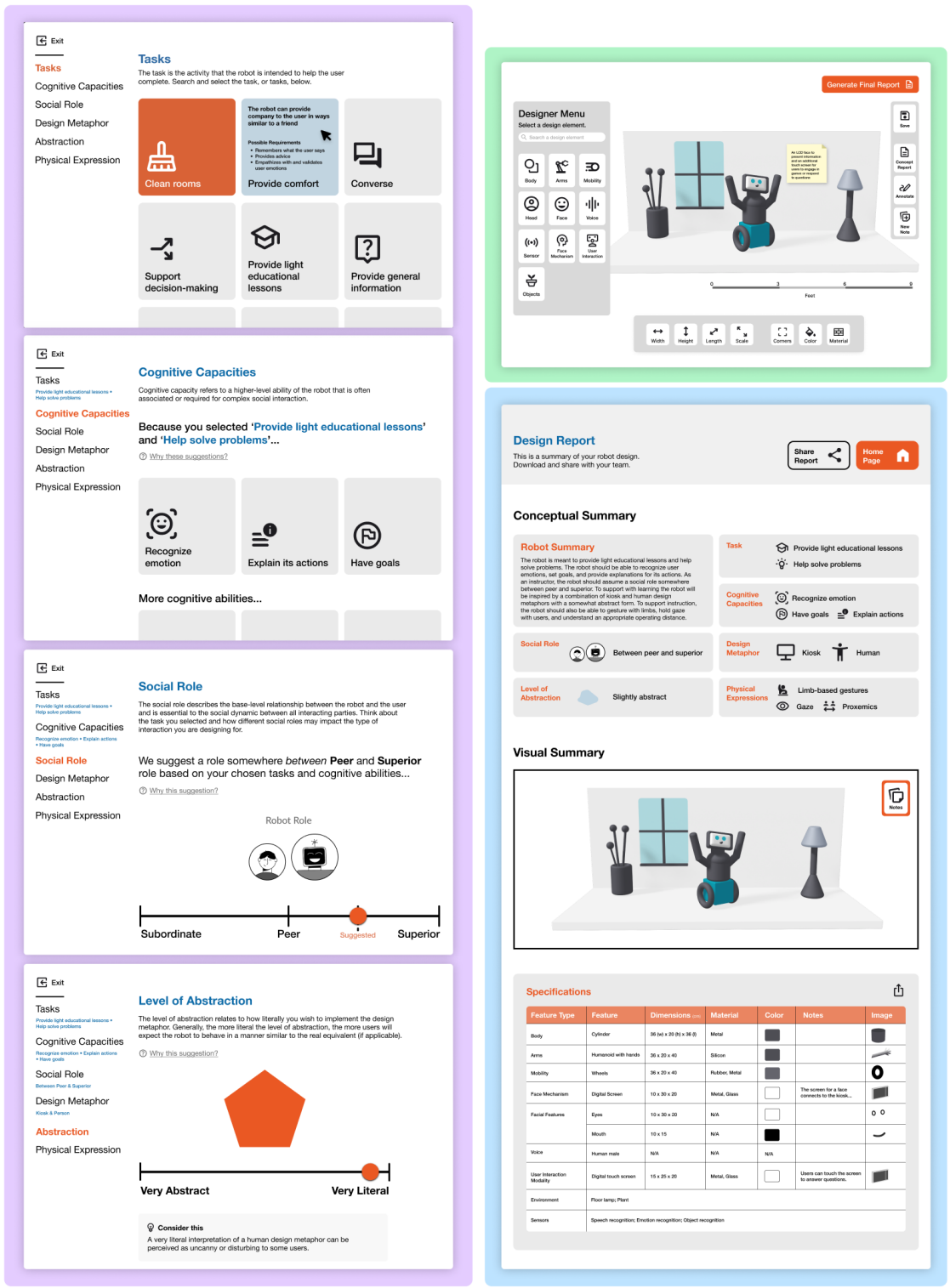


Figure 1: Major screens of the tool prototype, including the Conceptualization phase (left column highlighted in purple), the Visualization phase (Top of right column highlighted in green), and the final design report (Bottom of right column highlighted in blue).

3.3.3 Social Role Selection. Next, designers can choose the robot's social role embodied during its interactions with an end user. Designers interact with a sliding scale with 'subordinate' on the far left, 'peer' in the middle, and 'superior' on the far right. Users are guided with a suggested social role informed by trends observed in previous studies [8, 9].

3.3.4 Design Metaphor. Next, the Design Metaphor page is where the designer can select the inspiration for the robot's physical embodiment. To demonstrate, we have included nine design metaphors to choose from, with one being "freeform," which represents no strict metaphor. The metaphors include 'baby', 'butler', 'kiosk', 'person', 'dog', 'alien', 'animal', and 'bird'. These options were partly surveyed from an open-source database [9]. Each design metaphor card reveals the expectations and common features associated with that metaphor.

3.3.5 Level of Abstraction. The Abstraction page is dedicated to selecting an abstraction level that suits the robot considering all the design choices made until this point. The level of abstraction is represented by a sliding scale with a conceptual visual. To represent the abstraction level visually without priming designers with a specific interpretation of abstract designs, we introduced a shape analogy that transcends a specific robot embodiment. As the slider is moved from very abstract to very literal, the shape visualization transforms from amorphous to more rigid and structured.

3.3.6 Physical Expressions. The Physical Expressions page serves as the last step of the Conceptualization Phase. Here, the designer selects the types of physical expressions, or embodied cues, that apply to their robot. The design options are informed by embodied cues outlined in HRI literature as influential to complex social interaction [2, 8, 25].

3.3.7 Visualization Phase. The Visualization Phase introduces new interface and interaction design compared to the rest of the toolkit. Specifically, the interface more closely resembles a video game avatar builder or consumer product customization tool. On the left side of the screen is a hierarchical menu used to search through an array of physical features inspired in part by a list developed by Mahdi et al. (2022) [20] (see Appendix B.2). In addition, several core shapes and features were selected from a hands-on robot design kit used in an experiment to enable older adults to envision a useful social robot for themselves [14].

Another important feature of the Visualization phase is the adjustments toolbar. This toolbar enables the designer to make minute changes to the width, height, depth, scale, edge sharpness (i.e., how rounded an edge or corner is), color, and material of selected robot features. Material options include fur, metal, plastic, glass, silicon, and others.

Lastly, a menu located on the right side of the screen groups multiple actions such as a shortcut to the conceptual design report, the ability to add notes, and the ability to annotate with a drawing tool. These features enable the designer to keep track of their design rationale, support the documentation of different ideas, and allow designers to return to their designs without fear of forgetting important information.

3.3.8 Design Reports. The design reports are condensed summaries of the designer's choices and serve as a shareable deliverable that can support communication of design ideas and iterations with other designers, engineers, or end users. The designer can generate two reports throughout the process: once after completing the conceptualization phase and again after the visualization phase. The reports are characterized by consistent iconography and terminology that the designer will already be familiar with by the point of their generation. The final report also includes a specifications table that records the dimensions, color, material, and other adjustments made to each feature of the robot. The designer can also view any notes associated with the overall design or notes related to specific features of the robot.

4 PRELIMINARY EVALUATION

4.1 Interactive Expert Interviews

To begin evaluation of the prototype, we recruited two experts with research and practice experience in human computer interaction, social robot design, and interaction and product design. Expert one (E1) has experience in industrial and social robot behavior design and expert two (E2) has experience with social robot design through commercial and research projects. For each expert we conducted a two-hour, guided cognitive walk-through of the tool on Zoom. We required them to share a live visual of their screen while exploring the prototype from their computer. Experts were asked their thoughts on the tool's design, including how they perceived the content, organization, and usefulness of each feature and screen of tool. We also discussed how they might change the tool to further incorporate teamwork, collaboration, and participatory design methodology. Experts were compensated for their participation.

4.2 Emerging Themes

The initial qualitative analysis of our data highlights the following themes to improve the platform's design and usability.

4.2.1 Strengths and Potential Benefits.

- Support for interdisciplinary and participatory design:** The tool could facilitate design conversations among stakeholders - "...to have a place to start talking about the reasoning of my decision" (E2). E1 expressed that the tool could enable product managers, researchers, and educators with limited sketching abilities to generate numerous concepts and then select and iterate on ideas with their teams. Both participants also believed that the tool could help facilitate direct design with end users. E1 stated, "...you use this to walk through with someone who doesn't have design skills and try to get something they envisioned".
- Education:** E2 explored the idea of the tool's output as labels, similar to nutrition labels, on robots. They discussed that this could provide transparency to end users about the design process and the reasoning behind certain choices, enabling informed decisions when selecting a robot. Similarly, E1 discussed how the tool could support education, saying "I think it's a very, very good tool for an introduction of social robotics. It could be a very, very good tool for education, or

for people who want to get their hands on and get into social robotics."

- **Rapid concept generation:** Experts believe that the tool enables idea generation without substantial investment in learning software. E2 discussed how this platform can go beyond sketching, facilitating theoretical discussions about desired robot features, and can include the participation of future consumers in the design process.

4.2.2 Suggestions for Improvement. The following comments are presented in the order of which the associated screens were discussed during the interviews. For the tasks selection screen, E2 suggested to include example sensors a robot would need for the selected tasks. E1 recommended that the task options be categorized to help users find and select similar or related tasks. Referencing the cognitive capacities selection, E2 said, "I think it's hard for a robot in any task not to use any of these below". E2 also showed some concerns towards terms like "superior" or "subordinate" on the social role selection screen and suggested a scale based on knowledge levels for social roles. E1 shared that it should be clearer whose social role is being modified; the robot or the human. E2 also suggested exploring abstract design metaphors such as "sweetness or clouds" to convey desired interaction tones. E1 wished it was easier to distinguish metaphor options and understand "... what they are and what they mean". E2 expressed curiosity in exploring multiple abstraction levels for their design from this point onwards in the workflow. In addition to renaming the physical expressions screen to "Nonverbal Communication," E2 suggested incorporating a scale that considers the degrees of freedom of the robot. Both experts wanted to see a template or generated starting point in the builder based on the options they selected during the Configurations phase. E2 expressed the need for a clear connection between any recommended physical features and the selections made previously. E1 shared that the user interface could be consolidated to match other design and collaboration tools such as Adobe Suite or FigJam. One expert suggested incorporating a storyboard feature throughout the process to visualize selections and scenarios. This can help designers in gaining a holistic understanding of their design journey. The expert also proposed a design consideration to incorporate empty cards on all pages for personalization, brainstorming, and inclusion of new ideas and suggestions. Lastly, they expressed that the tool caters more to humanoid robot designs than novel abstract forms.

5 DISCUSSION AND CONCLUSION

Social robots are a promising and widely applicable technology that have yet to reach their full potential. Principles of human-centered and product design stress that the needs of users should be introduced early in the design process [10, 12, 15, 28]. However, there is a gap in available tools founded on human-robot interaction principles that support early-stage robot ideation and prototyping and enable designers with diverse levels of expertise to consider the complexities of social robots. The motivation of this study was to address this gap by introducing a digital design tool that walks designers through a structured, yet flexible social robot design process.

Our objectives were to create a platform that encourages designers to consider the physical, emotional, and behavioral aspects of their robot's design, enable rapid ideation and iteration, and support collaboration with development teams and end users. The outcome is a mid-fidelity prototype which was used to test and validate its effectiveness and inspire iterations for further designs.

Results from our preliminary evaluation with experts revealed strengths and opportunities for the tool. Experts generally acknowledged the gap in the social robot design process that the tool would address. They believe that the tool is a powerful conversation starter and capable of inspiring numerous robot design ideas. In future iterations, we wish to ensure the tool includes more abstract and non-humanoid robot features to choose from and better supports the exploration of novel robot forms. To do so, we will aim to incorporate more abstract design elements across the Conceptualization and Visualization phases. We also aim to continue our expert evaluation, further develop the Organize and Share phases, and conduct a second round of study targeting possible end users of the robot designs. The data gathered from this second study will supplement standard social robot design considerations with those specific to target end users. The findings from this stage will be combined with our existing research on potential application domains such as education, caregiving, and entertainment, helping to create use cases and scenarios that can be incorporated into the toolkit to guide the design process effectively.

We expect this tool to contribute to the field of social robot design by lowering the barrier to entry and by supporting the consideration of holistic design elements early in the design process. We also see this tool as generally supportive of interdisciplinary collaboration with other designers and the end users, such as facilitating participatory design, which is suggested to be an effective approach to creating social robots that align with end user needs [16, 30, 35] and utilizes techniques [30] that align well with the Conceptualization and Visualization phases of the proposed tool.

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A FIGURES

A.1 Webots Interface

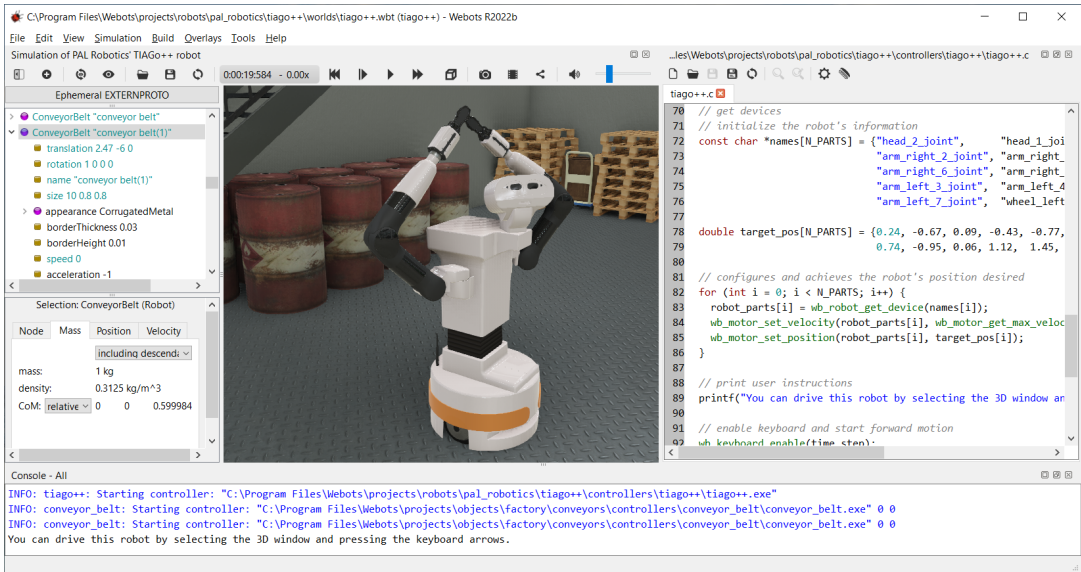


Figure 2: The Webots user interface from the Webots User Guide [19].

A.2 Proposed User Flow

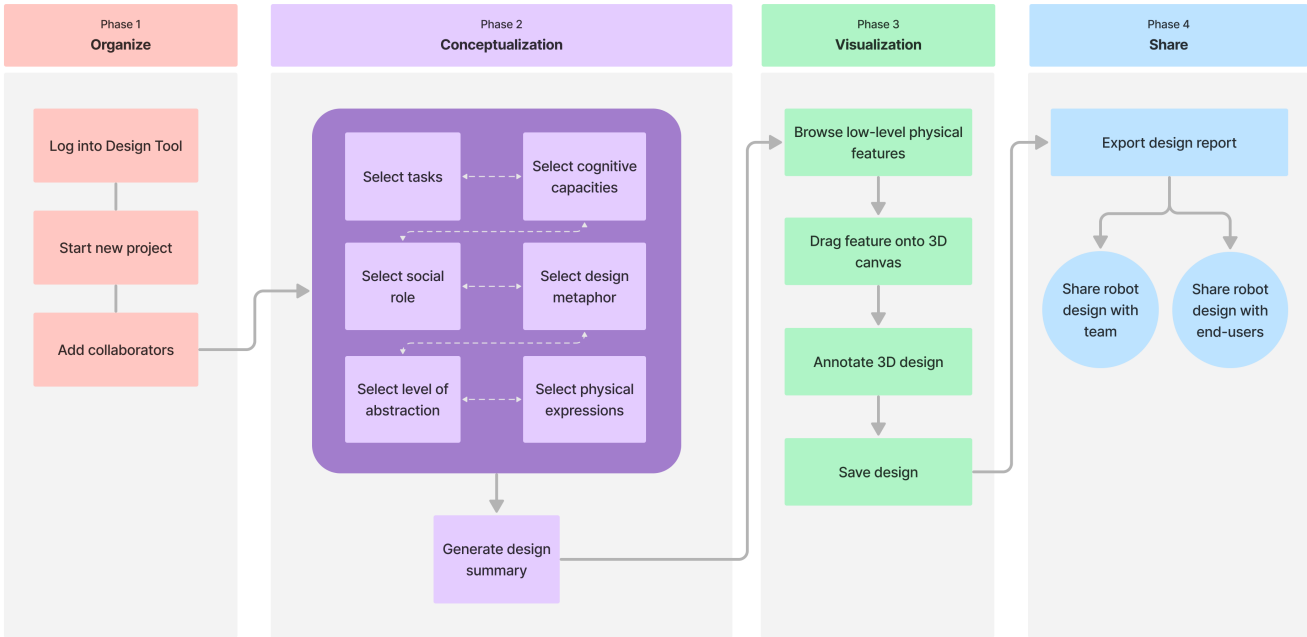


Figure 3: User flow of the proposed social robot design tool.

B TABLES

B.1 Design Requirements

Table 1: Design Requirements

Type	Requirement	Description
Creativity	Support Exploration	The tool must enable trying new ideas and reversing decisions, be self-revealing, unencumbered, and be fun to use [29].
Creativity	Low Threshold, High Ceiling, Wide Walls	The tool must be easy for novices to get started but support the creation of sophisticated, complete solutions. In addition, it must support and suggest a wide range of ideas [29].
Creativity	Support Collaboration	The tool must enable team members to share and contribute their unique knowledge [29].
Interface	Progressive Disclosure	Only provide information the user needs in the moment and in manageable chunks [33]
Interface	Aesthetic Consistency	Ensure consistency in the style and appearance of elements like font, color, and graphics to enhance recognition and expectations [17]
Interface	Iconic Representation	Use icons to make actions, objects, and concepts easier to find, recognize, and learn [17]
Interface	Internal Consistency	Ensure consistency in the grouping of aesthetic and functional elements to cultivate trust in the user [17]
Interface	Emphasis Mechanism	Bold words to draw attention to important information [3]

B.2 Available Robot Features

Table 2: Available Robot Features

Feature Category	Options
Body	Cube, Sphere, Cone, Rectangular Prism, Cylinder, Freeform.
Arms	Arm with humanoid hand, Arm with claw, Arm with animal paw, Wing
Mobility	Leg with humanoid foot, leg with paw, wheel
Head	Cube, Sphere, Cone, Rectangular prism, Cylinder, Freeform
Face Mechanism	Digital screen, Back projected, Fully actuated
Facial Features	Eyes, Mouth, Nose, Snout, Whiskers, Eyebrows, Ears
Voice	Human male, Human female, Human non-gendered, Dog bark, Cat meow, Animal sound (other)
User Interaction Modality	Physical buttons, Digital touch screen, voice command, remote control
Sensors	Speech recognition, Emotion recognition, Object recognition, Face recognition, Gesture recognition, Pressure sensor, Movement sensor, Heat sensor, Taste sensor, Smell sensor
Environment	Plant, Floor lamp, Window, Bed, Table, Rug